BUDAPESTI CORVINUS EGYETEM

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Seminar 1-2

Risk Management in Financial Institutions

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Risk Management in banking

Why do we concentrate on banking?

- Banking is a risky business
- Using depositors' money
- Role of banks: transfer funds to the most valuable projects that are beneficial to the economy
- **High leverage**
- Regulation – soundness – financial stability
- Risk management's focus is to manage the exposure to losses or risk and to protect the value of assets

Risks of banks

Credit risk

- Risk of not repaying the loan
- Failed delivery of loan agreement

Market risk

- Risk arising from market movements (interest rate, exchange rate etc)

Operational risk

- Risk arising from human and system failures, external events and process deficiencies

Liquidity risk

- Lack of ability to pay debts
- Risk arising from the different maturity of assets (loans/receivables) and liabilities (deposits/funds)

Systemic risk

Financial system instability, potentially catastrophic, caused or exacerbated by idiosyncratic events or conditions in financial intermediaries

Connection of risks

1. Credit Risk (sub-prime crisis) → 2. Counterparty risk (Lehman) → 3. Global liquidity Risk → 4. Credit and market risk → 5. Budget expansion, vulnerable growth → 6. Country Risk

History of bank regulation

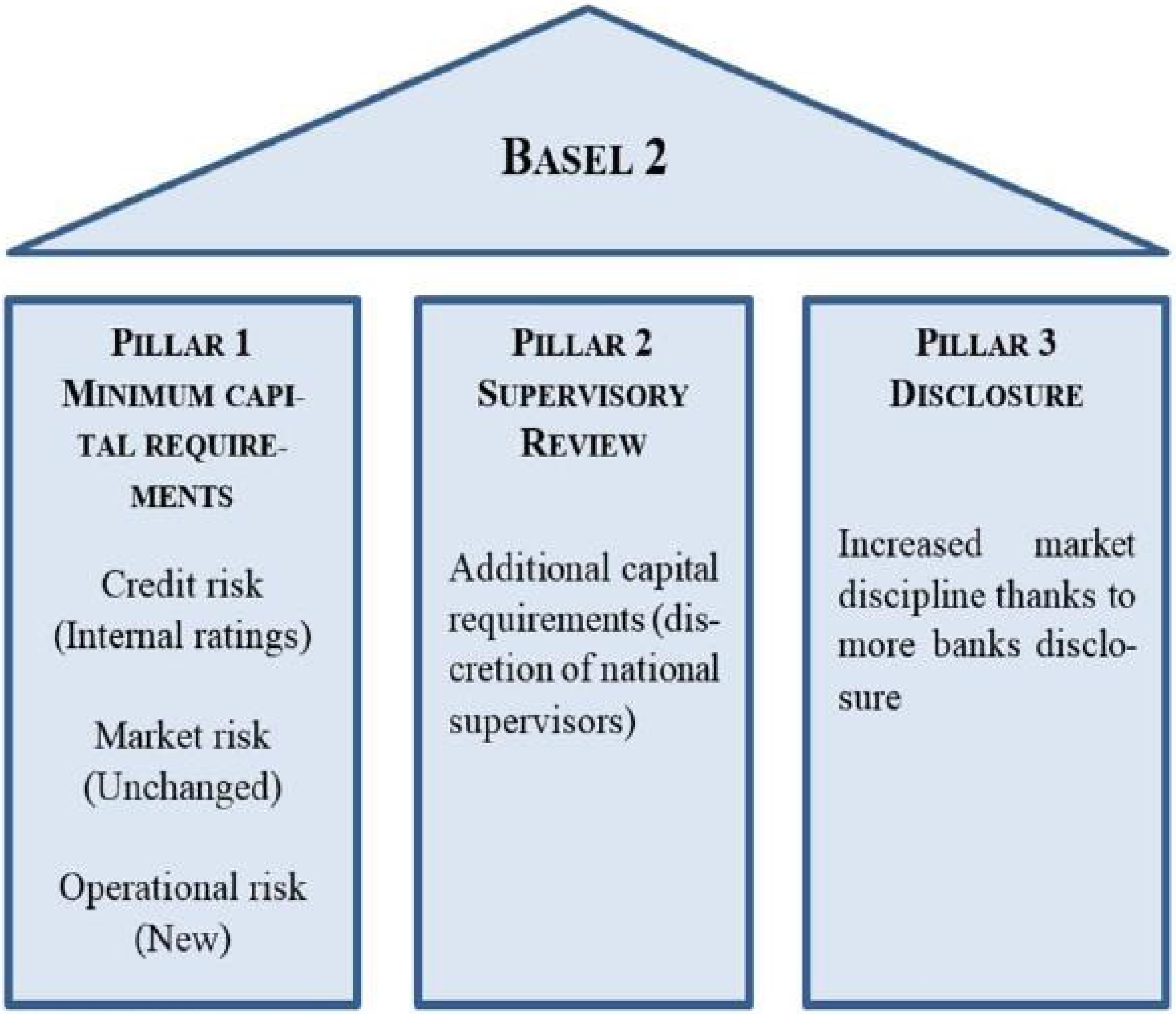
- 1974: Basel Committee on Banking Supervision
- 1988: BIS Accord (Basel I.)
- 1996: Amendment to Basel I.
- 2004: Basel II.
  - CRD I. 2006; CRD II. 2009
- 2009: Basel II.5
  - CRD III. 2010
- 2011: Basel III.
  - CRD IV: Directive 2013/36/EU; CRR: Regulation (EU) No 575/2013

Basel standard

- Basel I. – credit risk
- 1996: Amendment – market risk of the trading book
- 2004: Basel II. – operational risk
- 2009: Basel II.5 – market risk revisited
- Basel III. – liquidity risk, systemic risk

Capital calculation;

Requirements toward risk management system


BASEL 2
PILLAR 1
MINIMUM CAPITAL REQUIREMENTS
Credit risk
(Internal ratings)
Market risk
(Unchanged)
Operational risk
(New)
PILLAR 2
SUPERVISORY REVIEW
Additional capital requirements (discretion of national supervisors)
PILLAR 3
DISCLOSURE
Increased market discipline thanks to more banks disclosure

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Risk

Risk versus Uncertainty

- **Risk:** the outcome is unknown (random), but the distribution is known (we know the possible outcomes and their probabilities).
- **Uncertainty:** the probabilities or even the possible outcomes are unknown.
- **Risk measure:** is a function that orders a value to a loss distribution (that is the capital to be added to a portfolio to achieve the required risk level).

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Intro – Risk

- Trade off between risk and expected return
  - The greater the risks, the higher the return
- Expected return: weighted average of possible returns
  - Weight reflects probabilities
- 100.000 USD invest for 1 year
  - Treasury bills yield 5%
  - Stock with the outcomes represented in the table
- Best-case scenario: 30% return, + 30.000 USD
- Worst-case scenario: -20% loss, -20.000 USD

Probability	Return
0,6	+30%
0,4	-20%

$$\boxed{0{,}6 \times 0{,}3 + 0{,}4 \times (-0{,}2) = 0{,}1 \rightarrow 10\%}$$

Measure of risk

<u>Value at Risk (VAR):</u> The maximum loss that a portfolio may incur in a given period, with a given probability (significance level), under normal market conditions

<u>Expected shortfall (ES):</u> The expected loss we suffer in the worst specified percentage of cases

Value at risk (VaR)

The maximal loss, we can suffer under a given period and significance level under normal market circumstances.

VaR (1 day, 99%) = USD 1 million

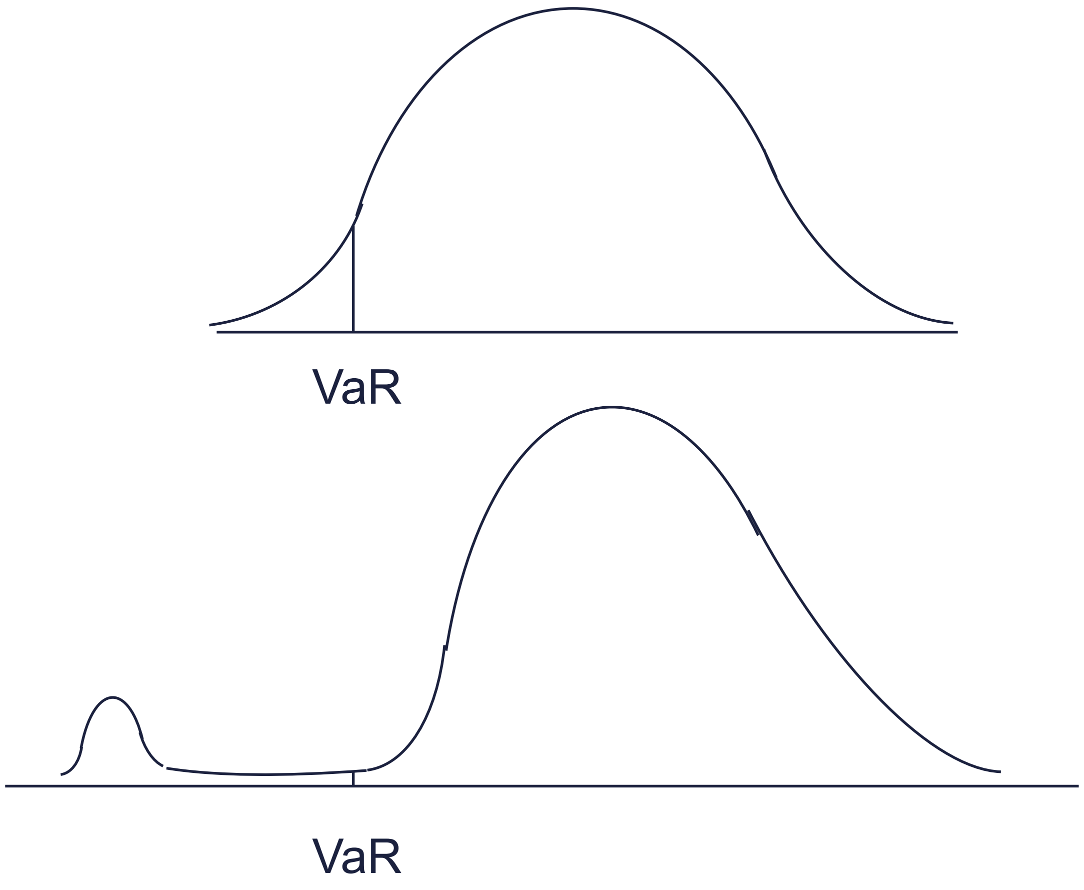
We will not loose more than USD 1 million

with a probability of 99%

"How bad can things get?"

Expected shortfall (ES)

- Expected shortfall (ES) is the expected loss given that the loss is greater than the VaR level (also called C-VaR and Tail Loss).
- ES is coherent, VaR is not (diversification)
- “If things do get bad, what is the expected loss?”



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VAR Multiplier

$$VAR = \mu + \sigma * N^{-1}(1-\alpha)$$

Szig	VaR-multiplier	ES-multiplier
95%	- 1,6449	2,0627
99%	- 2,3263	2,6652

ES Multiplier

$$ES = \mu - \sigma \frac{e^{-Y^2/2}}{\sqrt{2\pi}(1-\alpha)}$$

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What de we know about risk and return

Formulas

$r_t = \frac{S_t}{S_{t-1}} - 1$

$y_t = \ln\left(\frac{S_t}{S_{t-1}}\right)$

$r_p = \sum_{i=1}^{n} w_i \cdot r_i$

$\sigma_p^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} cov_{ij} \cdot w_i \cdot w_j$

$dS = \mu S dt + \sigma S dw$

$dY = (\mu - \frac{\sigma^2}{2}) dt + \sigma dw$

$\Delta Y = (\mu - \frac{\sigma^2}{2}) \Delta t + \sigma \Delta w$

Price and return

$y \sim N(\left(\mu - \frac{1}{2}\sigma^2\right) t \,;\, \sigma\sqrt{t})$

$S_T = S_0 e^{y}$

$E(S_T) = S_0 e^{\mu T}$

$\sigma(S_T) = S_0 e^{\mu T} \sqrt{e^{\sigma^2 T} - 1}$

Thank you for your attention!

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